

Emotion pipeline

GROUP 4

Contents

- Business
- Data
- Models
- Full pipeline
- Ethical considerations
- Future directions

Business

- Our understanding of the problem – LLM's
- The goal – cheaper and revolutionary
- Our project – New pipeline integrations

The data



Emotion analysis



Go emotions (58k samples)



Additional data added



Machine translation



Open subtitles NL - EN



200k sentences



Problems



Test data

Speech to text



Assembly AI



Whisper



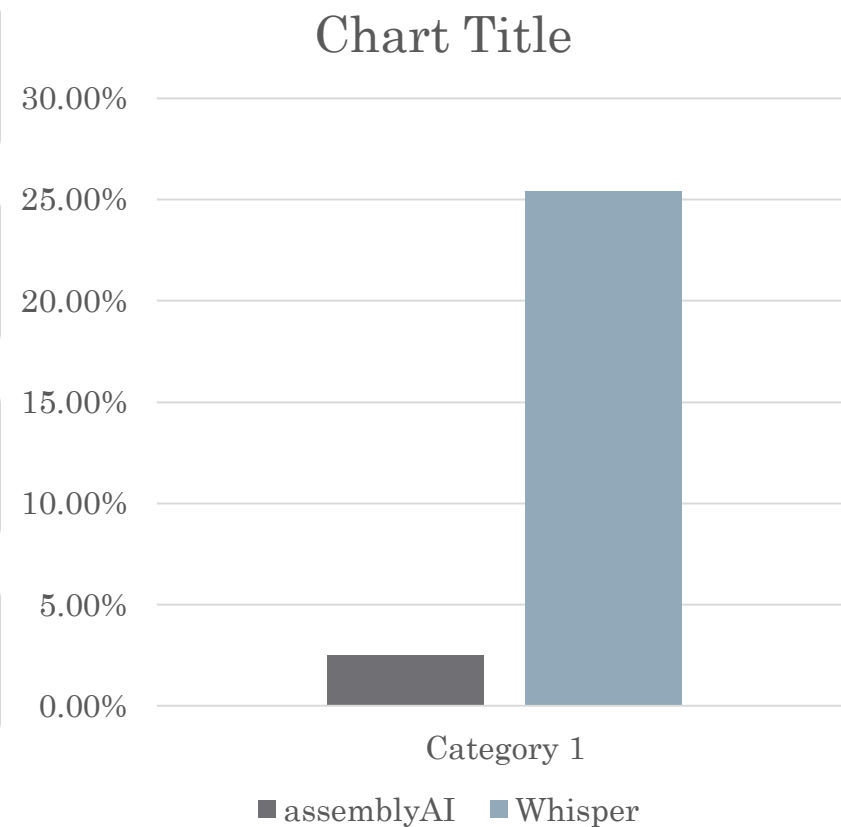
Problems



Tradeoff



Results

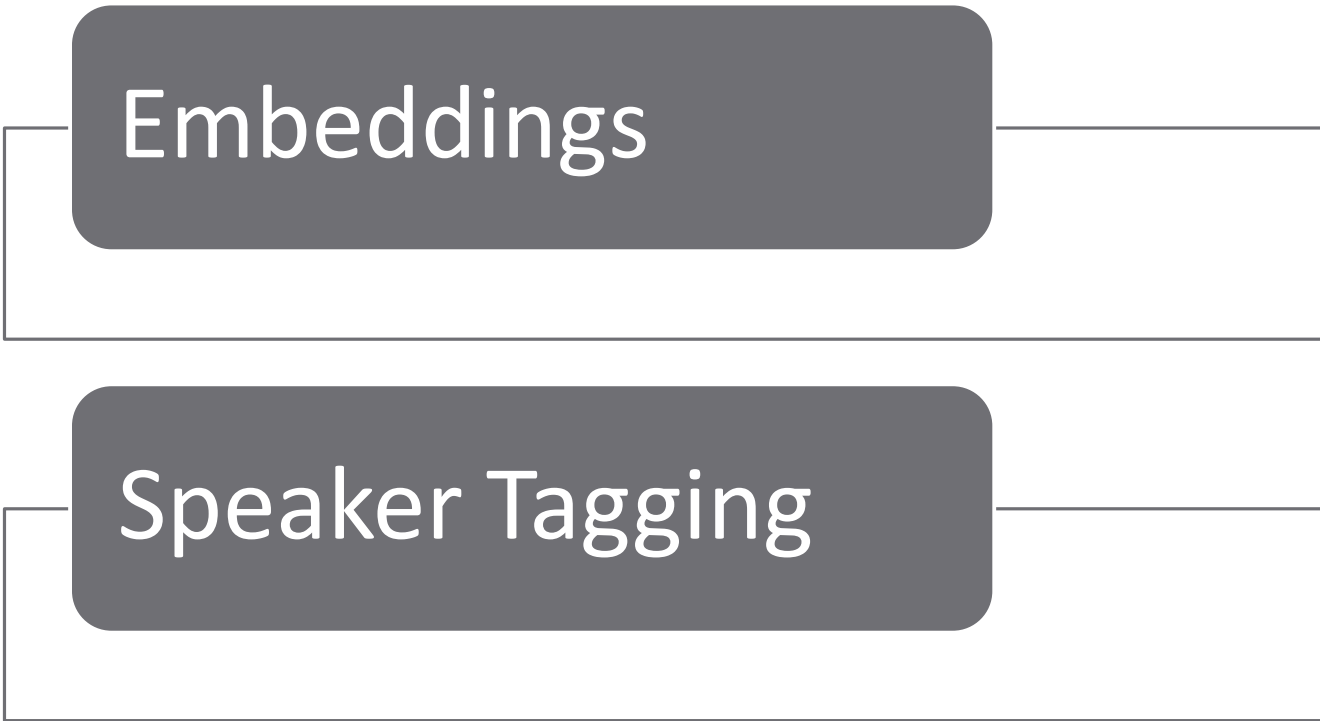


Features

TF-IDF

POS-Tagging

Features



Embeddings

Speaker Tagging

Models

Name	Evaluation	Pro's	Con's
Logistic Regression	Weighted F1 0.46 Minority classes rarely classified	Fast & Easy	Doesn't perform well on non-linear data
Naïve Bayes	Weighted F1 0.49 Completely failed to classify minority classes	Works well on small datasets	Doesn't capture relationship between words
LSTM	Weighted F1 0.55 Completely failed to classify minority classes	Word order Long term dependencies	Prone to overfitting Computationally heavy to train

Transformer

Pros:

- Contextual understanding

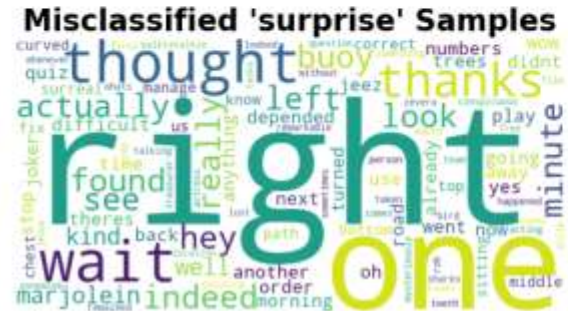
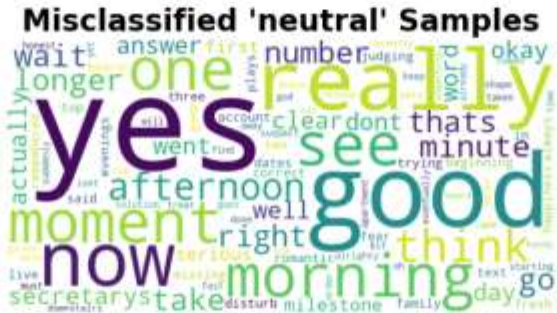
Cons:

- Computationally heavy to train
- Low explainability

- Roberta-base-go-emotions
- Test Weighted F1 Score: 0.77
- Not able to effectively capture minority classes

Transformer Results

- Translation errors
- Short sentences contribute a lot of noise to the neutral
- Most confused classes Happiness and neutral



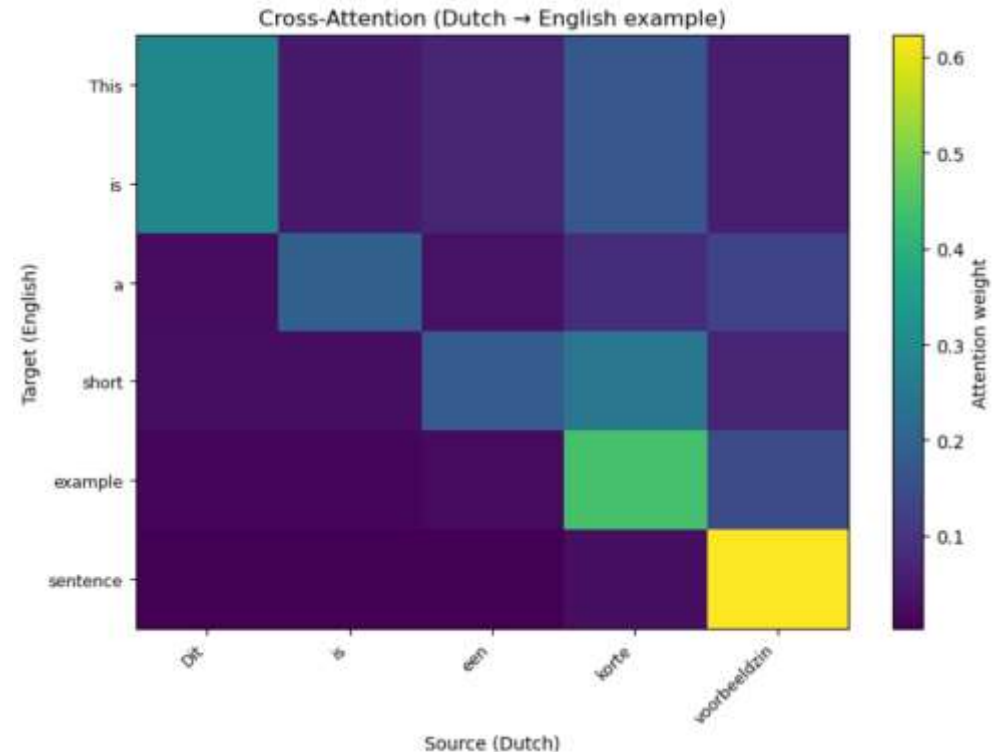
Machine Translation

Data preprocessing and
architecture (enc – dec)

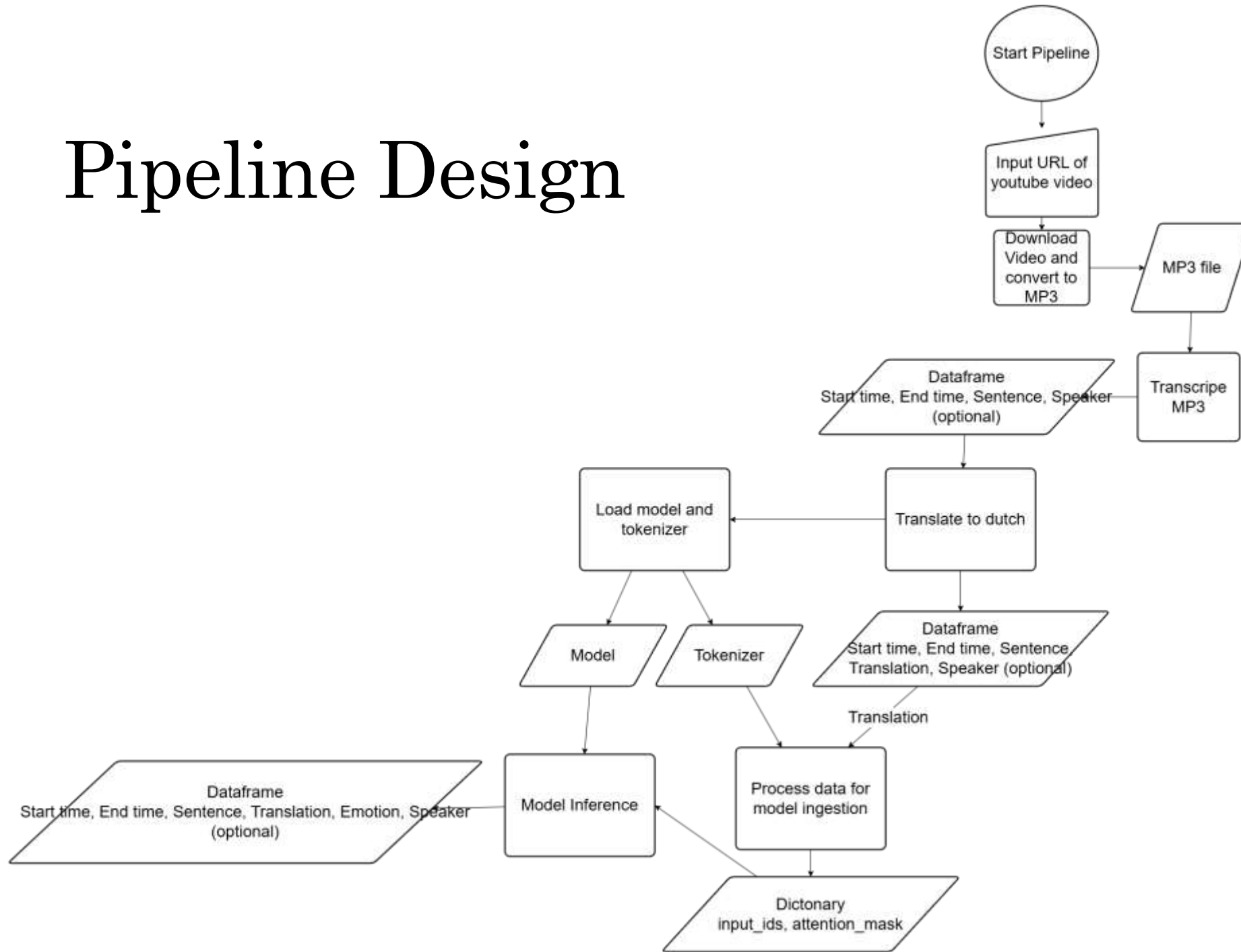
Training

Results with checking
and BLEU

Future direction in the
pipeline



Pipeline Design



Pipeline Limitations

- Transcription accuracy VS Computational time
- Low recall and accuracy for minority classes
- Video download with only works on youtube
- Feature extraction not integrated

Ethical considerations



Data privacy



Emotional
profiling



Representation

Future directions

Data limitations

Full integration

Model optimization